

Program: Diploma Cyber Forensics and Information Security	
Course Code: 4287	Course Title: Programming Lab through Python
Semester: 4	Credits: 1.5
Course Category: Program Core Course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To write, test, and debug simple Python programs.
- To implement Python programs with conditionals and loops.
- Use functions for structuring Python programs.
- Represent compound data using Python lists, tuples, dictionaries.
- Read and write data from/to files in Python

Course Prerequisites:

Topic /Description	Course Code	Course Title	Semester
Knowledge of Computer Basics		Introduction to IT systems lab	I
Basic Knowledge of problem solving		Problem Solving and Programming	II

Course Outcomes

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Solve problems using basic programming concepts and control structures	9	Applying
CO2	Use lists, tuples and dictionaries for representing complex data and develop user defined functions to implement modular programming and string handling.	13	Applying
CO3	Illustrate exception handling and object oriented programming concepts such as constructor, encapsulation, inheritance and polymorphism.	13	Applying

CO4	Develop programs using files and database connectivity	10	Applying
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CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3	3			2
CO2	3		3	3			2
CO3	3		3	3			2
CO4	3		3	3			2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Solve problems using basic programming concepts and control structures		
MO1	Develop programs to evaluate simple mathematical expressions	3	Applying
MO2	Make use of selection control structures to solve problems	3	Applying
MO3	Develop programs using loop control structures	4	Applying
CO2	Use lists, tuples and dictionaries for representing complex data and develop user defined functions to implement modular programming and string handling.		
MO1	Implement real-time/technical applications using Lists, Tuples and Dictionaries	3	Applying
MO2	Develop programs using user defined functions	5	Applying
MO3	Develop programs to handle strings	5	Applying
CO3	Illustrate exception handling and object oriented programming concepts such as constructor, encapsulation, inheritance and polymorphism.		
MO1	Construct real-time/technical applications using Exception handling. (divide by zero error, voter's age validity, student mark range validation)	5	Applying
MO2	Develop programs using class and objects and demonstrate the concept of constructor and inheritance.	5	Applying

CO4	Develop programs using files and database connectivity		
MO1	Implementing real-time/technical applications using File handling. (copy from one file to another, word count, longest word)	4	Applying
MO2	Implement python programs with database connectivity	3	Applying

Text / Reference:

T/R	Book Title / Author
T1	John M. Zelle, Python Programming: An Introduction to Computer Science.
T2	Allen B. Downey, “Think Python : How to Think like a Computer Scientist”, 2nd Edition, O’Reilly Publishers, 2016.
R1	Paul Deitel and Harvey Deitel, “Python for Programmers”, Pearson Education, 1st Edition, 2021.

Online Resources:

SI No	Website/Link
1	https://archive.org/details/MIT6.00SCS11
2	http://coursera.org/course/pythonlearn